

The Riverlabs Wari Water Quality Buoy is a compact, self-righting floating logger for continuous in-situ monitoring of water quality in rivers, lakes and reservoirs. Each buoy carries turbidity, water temperature and electrical conductivity sensors, and is designed to be deployed individually or as a low-cost swarm across a catchment.

The buoy is built on the same low-power ATmega328P logger platform as the Wari water level loggers, combined with a high-precision real-time clock, 512 Kbit EEPROM and 64 Mbit flash memory. The hull is 3D-printed ABS with an epoxy coating and a closed-cell neoprene seal, with a keel weight to keep the buoy upright and the sensors submerged. A single hitching point at the bow allows mooring to a fixed structure or anchor.

Firmware is fully open source and programmed in the Arduino environment, so the sampling interval can be adapted to each project. Data is stored internally and retrieved via an FTDI cable, or transmitted in near real time with the optional LoRaWAN radio module. Every buoy is individually calibrated against laboratory standards before dispatch.



Technical Specifications

Parameter	Specification
Turbidity	0 – 4000 NTU (resolution 5 NTU, accuracy ± 10 %)
Temperature	$-55 - 315^{\circ}\text{C}$ (resolution 0.1°C , accuracy ± 1 %)
Conductivity (FEC)	0 – 2000 $\mu\text{S/cm}$ (resolution $2 \mu\text{S/cm}$, accuracy ± 10 %)
Sampling Interval	User configurable
Microprocessor	ATmega328P (Arduino compatible, MiniCore bootloader)
Memory	512 Kbit EEPROM + 64 Mbit internal flash (FTDI read-out)
Clock	High-precision real-time clock with coin-cell backup
Power Supply	1 x 3.7 V Li-ion battery (14500)
Hull & Seal	Epoxy-coated ABS, self-righting with keel weight; closed-cell neoprene gasket
Mooring	Single hitching point
Weight	Approx. 300 g

Key Features

- Three water quality parameters in one compact floating logger
- Self-righting hull keeps sensors submerged and antenna above water
- Low cost – designed for swarm deployments across a catchment
- Fully customisable open-source firmware
- Individually calibrated against laboratory standards
- Simple maintenance – wipe clean and download data monthly
- **Measured parameters:** turbidity, water temperature, electrical conductivity, battery voltage

Available Options

- Internal storage only
- LoRaWAN radio module – including antenna
- Additional sensors (e.g. optical pH) under development – contact us